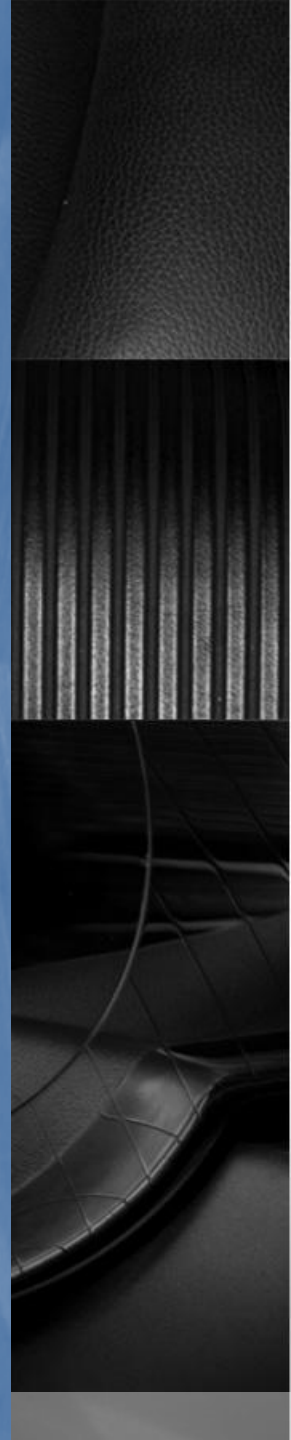
COMP4431 Artificial Intelligence

Heuristic Search

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Partly based on Prof. Liu Slides



Agenda

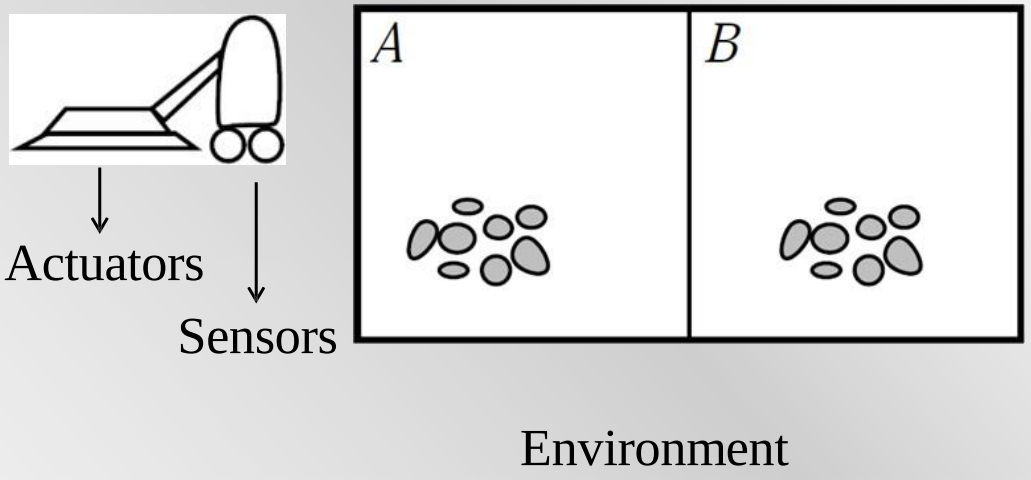
- Rational agent
- Search agent
  - Uninformed search and informed search
  - A* search
  - Heuristics
  - Domination of heuristic functions
- Technological trends of agent

Agenda

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- Search agent
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- Technological trends of agent

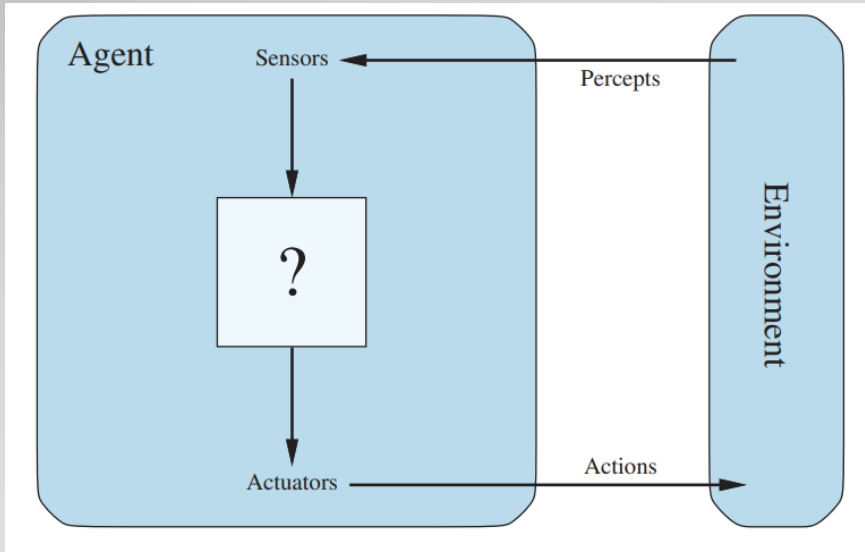
Agent

- Agents in artificial intelligence
  - Agents denotes artificial entities that are capable of perceiving their surroundings using sensors, making decisions, and then taking actions in response using actuators
- Example
  - Vacuum-Cleaner
- How to perceives the environment
  - which room, clean or dirty
- How to make the decision
  - move right or left, suck the dust
- How to make the action



Rational Agent

- Agent
  - An agent is an entity that perceives and act with the environment
- Rational Agent
  - Agent that does the **right thing!**
  - Acts to achieve the best outcome
  - Different environment can have different agent to adopt it
  - For any given class of environments and tasks, we seek the agent (or class of agents) with the best performance



Rational Agent

- Characteristics of rational agent
  - Distinct from omniscience
  - An agent is autonomous if its behavior is determined by its own experience
  - Agents can perform actions in order to modify future percepts so as to obtain useful information

A real-life rational agent analogy

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Agenda

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Scenario

- Students visit scientific park
  - Use intelligent information system to aid arrangement
- Real world problem
  - Students are in building A
  - Students want to go to building G
  - Find a path from building A to building G
  - Find a shortest path from building A to building G
  - Find a shortest path from any building to another building



Uninformed Search

- Uninformed search looked through search space for **all possible solutions** of the problem without having any additional knowledge about search space
  - In the path-finding problem, if you are not given the full map of the location, you can only explore your neighbors repeatedly and see if you arrive

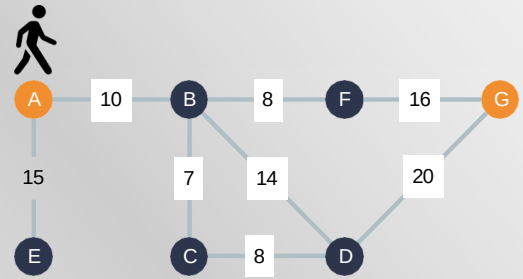


Informed Search

- But informed search algorithm contains an array of knowledge such as how far we are from the goal, path cost, how to reach to goal node, etc
- This helps agents to explore less to the search space and find more efficiently the goal node

Represent the Map using Graph

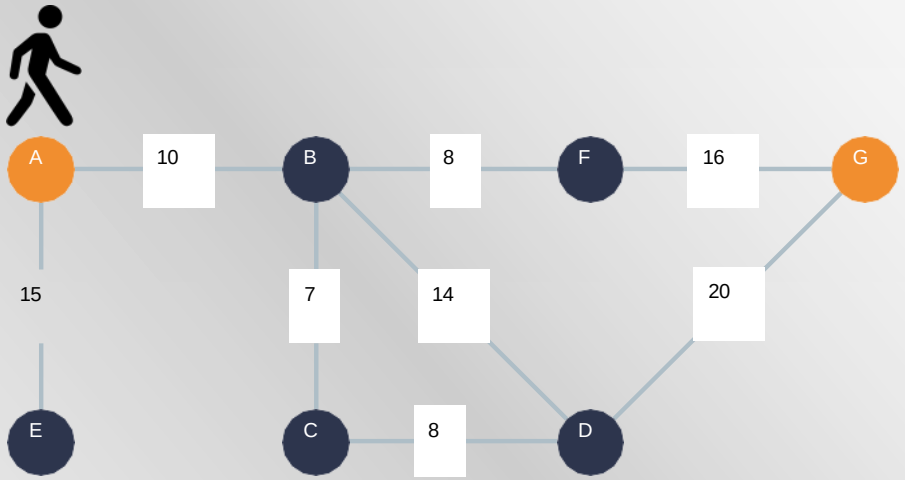
- Abstract the map to the graph
  - Ignore the detailed information
- Node
  - Location of the building
- Edge
  - The path connecting two buildings
- Value
  - The time walking from one building to another





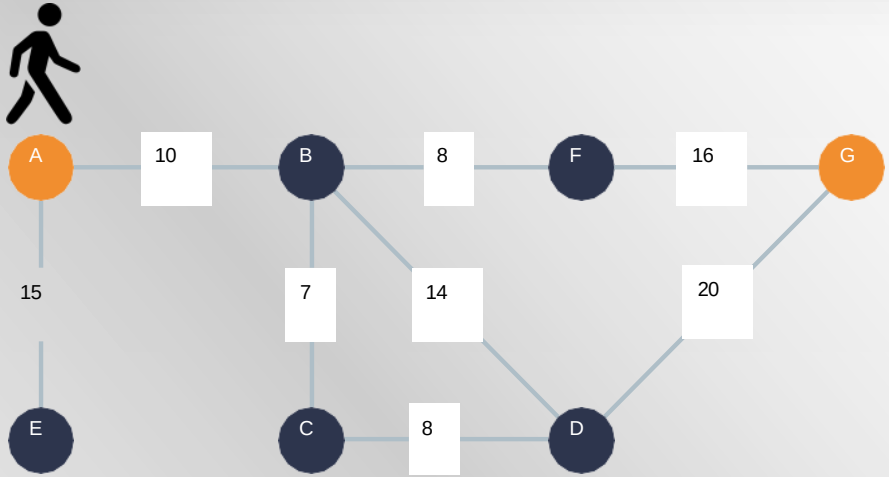
Solve the Problem using Rational Agent

- How to perceive the environment
  - Which kind of information should be collected?
  - Which kind of information could be collected?
  - Which kind of information should be kept?
  - Which kind of information could be kept?
- How to make the action
  - Which kind of action could be taken?
  - Which kind of action should be made?
- How to optimize the performance?
  - Could we find the solution?
  - Could we find the optimal solution?
- How to make the decision
  - Check the target
  - Check the cost



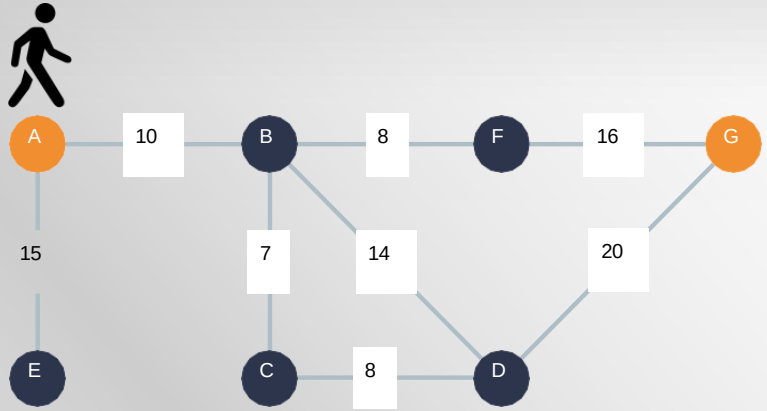
Problem Formulation

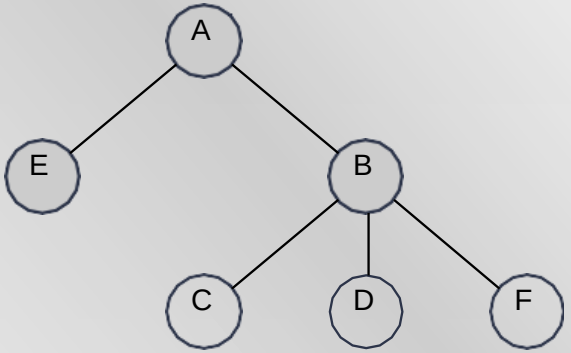
- Perception
  - State: In(B)
  - Initial state: In(A)
  - Goal state: In(G)
- Action
  - Decision result Go(B)
  - Solution: {Go(B),Go(F),Go(G)}
- Transition model
  - Action leads to the state transtition
  - Result (In(A), Go(B)) =In(B)
- Cost
  - If optimize the performance
  - c(In(A), {Go(B)}, In(B)) =10
  - The sum of the costs of the individual actions along the path
  - c(In(A), {Go(B), Go(F)}, In(F)) =10 +8 = 18



Any Existing Methods to Model the Problem

- Solution
  - An action sequence
- Search algorithm
  - Considering various possible action sequences
- Search tree
  - Nodes: states in the state space
  - Root: initial state
  - Branches: actions
- Expanding the current state
  - Apply each legal action to the current state, thereby generating a new set of states
  - Add branches from the parent node leading to child nodes
- Essence of search
  - Following up one option now and putting the others aside for later
  - In case the first choice does not lead to a solution





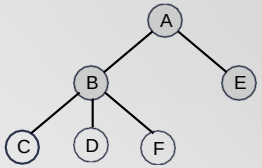
Breadth-first (BFS) & Depth-first Search (DFS)

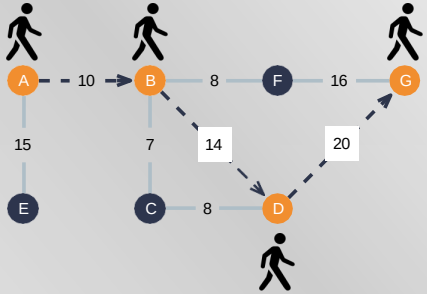
- Both are uninformed search strategies
  - Not necessary to solve path-finding
- Breadth-First Search (BFS):
  - Explores graph level by level, visiting all neighbors before moving deeper
  - Uses a queue data structure when implemented
- Example use: Finding shortest path, web crawling
- Depth-First Search (DFS):
  - Explores as far as possible along each branch before backtracking
  - Uses a stack or recursion when implemented
- Example use: Maze solving, detecting cycles

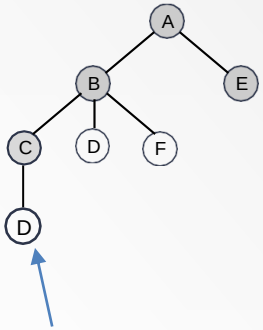
Breadth-first Search

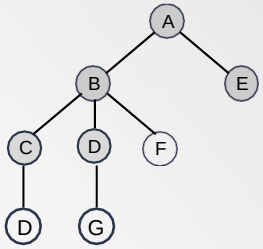
- Expand the shallowest unexpanded node
  - Cost =10+14+20=44

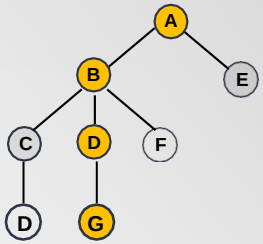








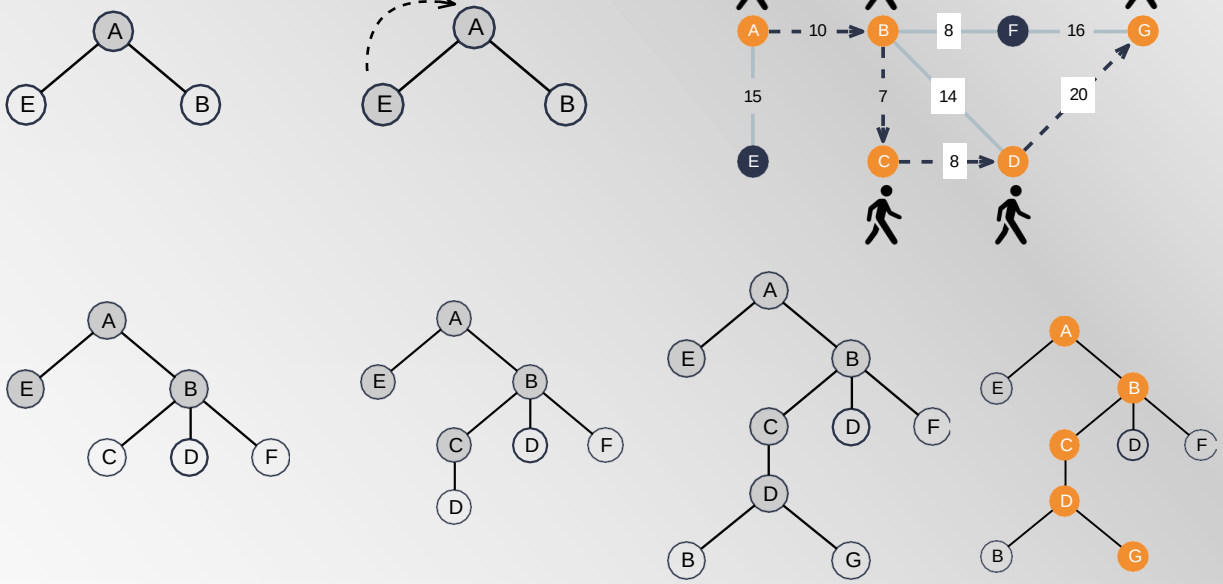




Here, as node D is not yet expended, so will still include it in consideration. (This may be different from some other textbooks)

Depth-first Search

- Expand the deepest unexpanded node
  - Cost =10+7+8+20=45



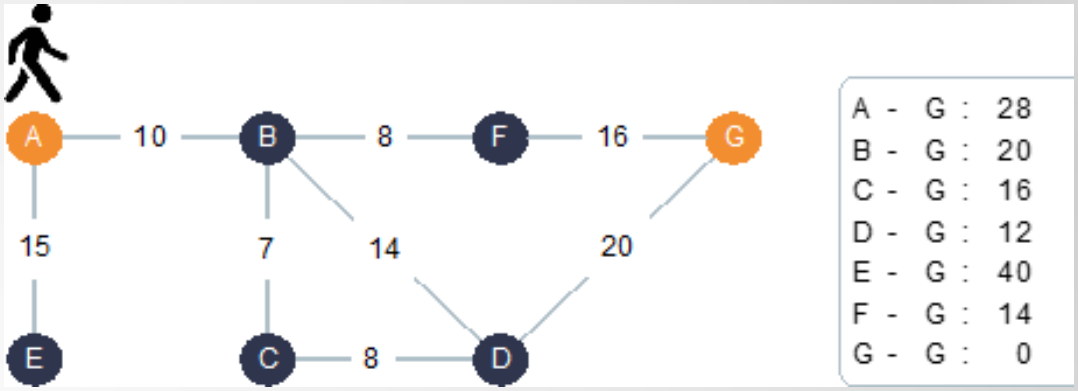
The Search Tree

- Observe the search tree which represent the state space
  - i.e. all possible states to reach our goal
- if the problem is large, the search tree can even larger!!!
- Search will take more time.
- Search strategy is important!



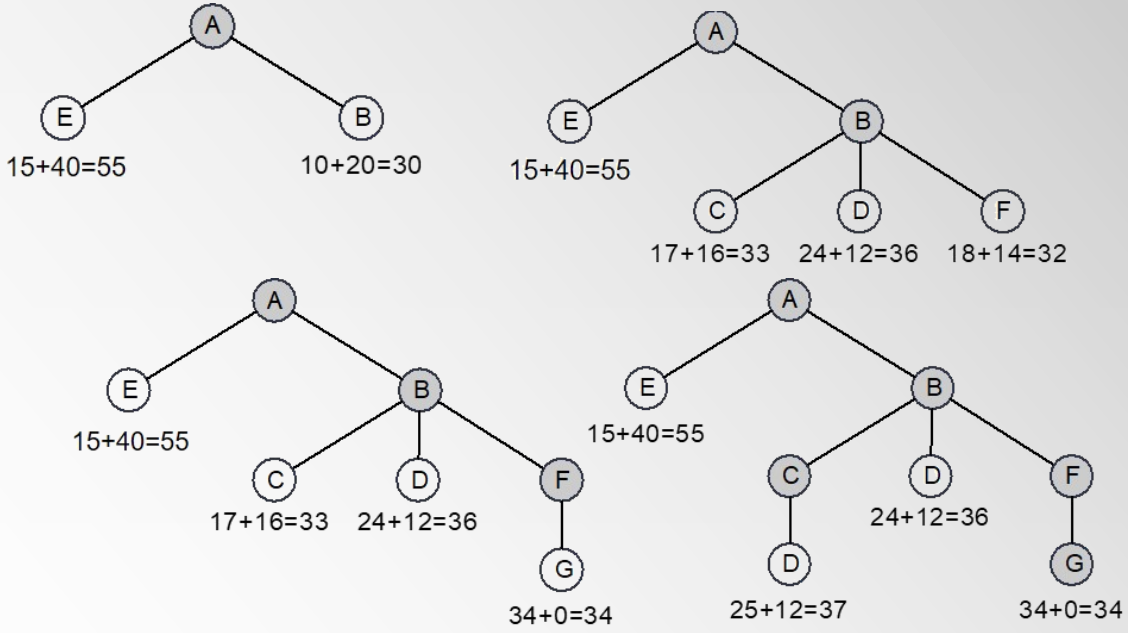
Uninformed Search and Informed Search

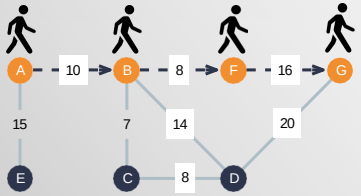
- Uninformed search
  - The strategies have no additional information about states beyond that provided in the problem definition
- Informed search
  - The strategies know whether one non-goal node is “more promising” than another
  - Heuristic information represents the relative suitability of a solution among its peers based on the intuition, experience or common sense

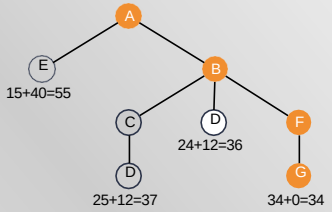


A* Search

- Evaluation function: f(n) =g(n) +h(n).
  - g(n): the cost so far to reach the node n
  - h(n): the estimated cost to goal from the node n
  - f(n): the estimated total cost of path through the node n to goal



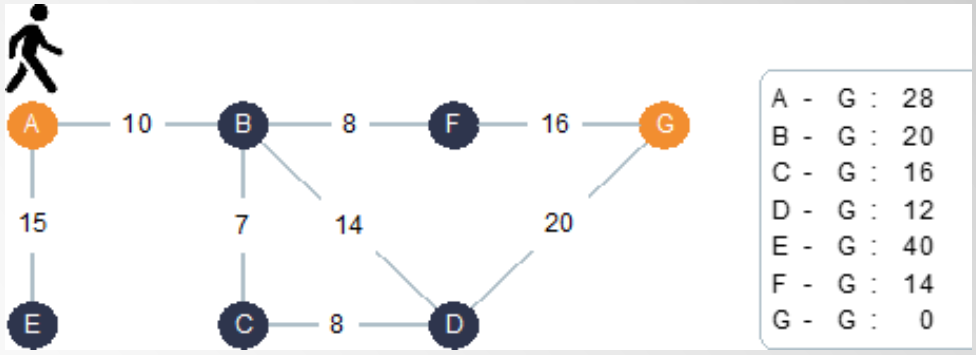




A	-	G :	28
B	-	G :	20
C	-	G :	16
D	-	G :	12
E	-	G :	40
F	-	G :	14
G	-	G :	0

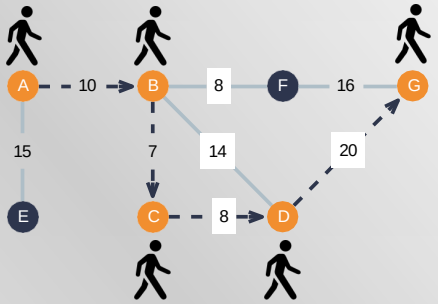
Heuristics

- Heuristics
  - A problem-solving method
  - Uses shortcuts to produce good-enough solutions given a limited time
  - Solution may not necessarily be optimal solution
- Heuristic function in search algorithms
  - A function that ranks alternatives based on available information
- An admissible heuristic never overestimates the cost to reach the goal
  - A heuristic h(n) is admissible if for every node n, h(n) ≤h*(n), where h*(n) is the true cost to reach the goal state from n



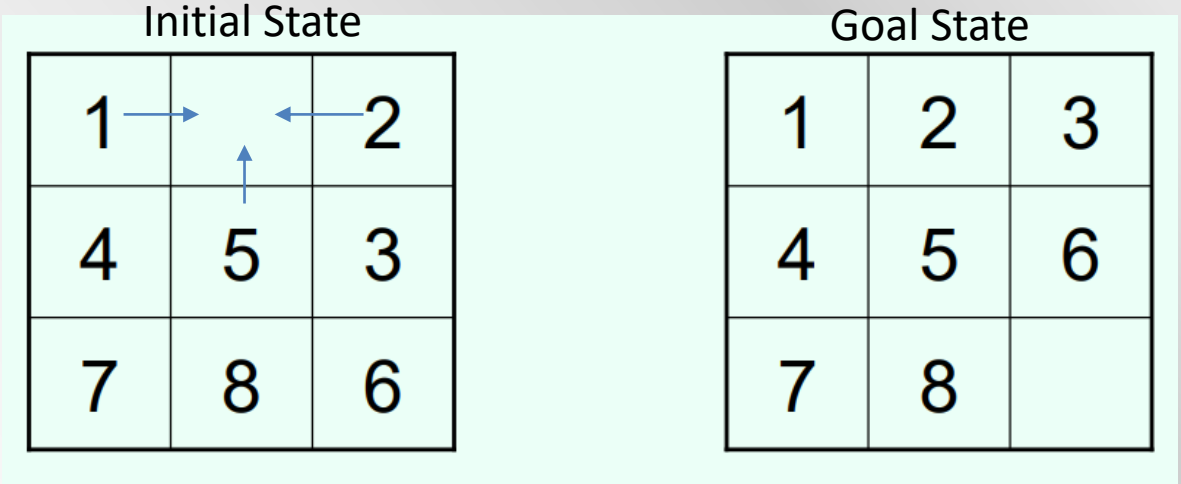
Search Strategy

- Search strategy
  - Pick the order of node expansion
- Strategies are evaluated along the following dimensions:
  - completeness: does it always find a solution if one exists?
  - time complexity: number of nodes generated
  - space complexity: maximum number of nodes in memory
  - optimality: does it always find a least-cost solution?
- Uninformed search algorithms
  - Breadth-first search
  - Depth-first search
- Informed search algorithms
  - A* search



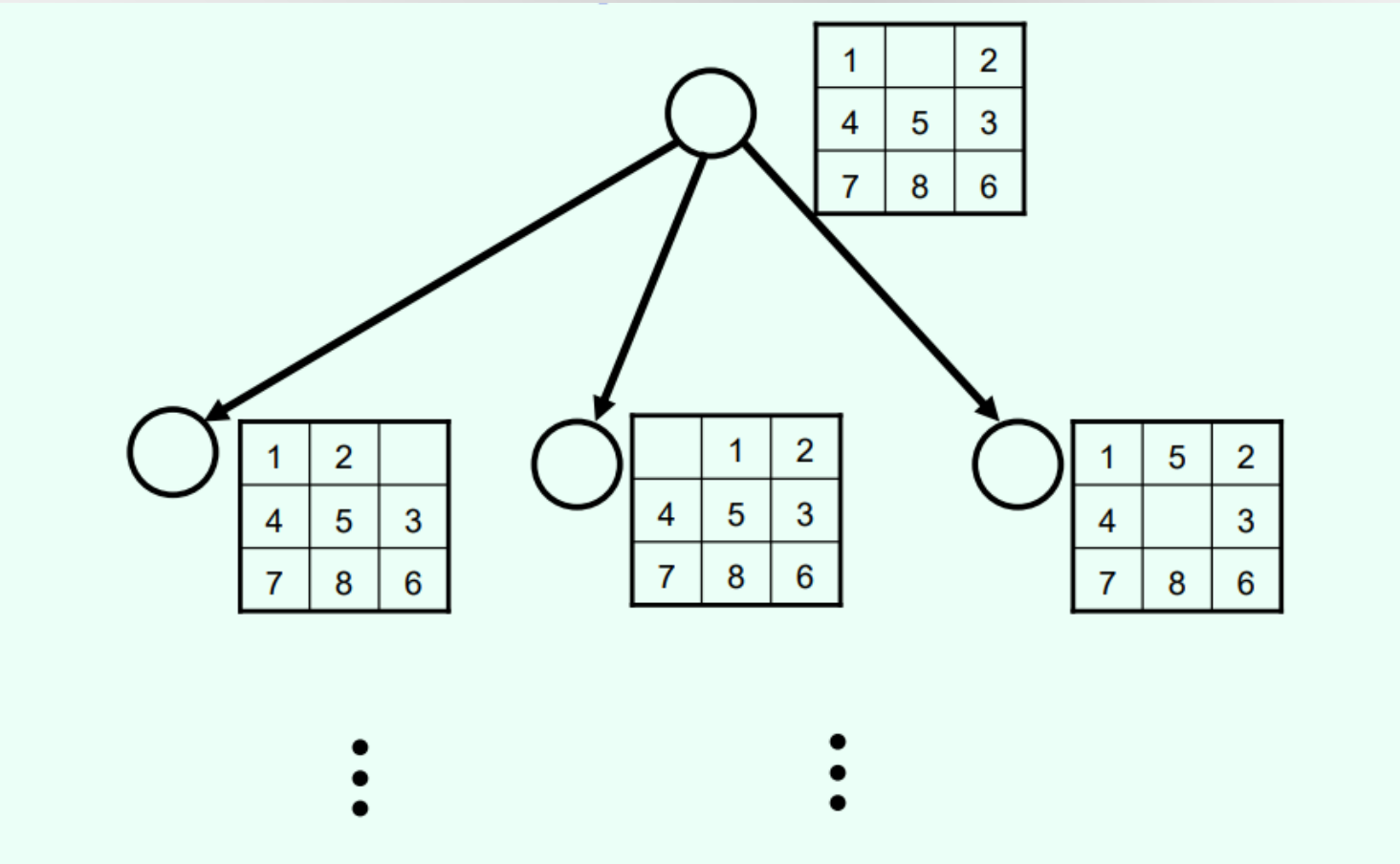
8-puzzle Problem

- Apart from path finding problem, searching can apply to other problem, e.g. 8-puzzle/sliding-tile puzzle
- The empty space can only move in four directions viz.,
  - 1. Up
    2.Down
    3. Right or
    4. Left



8-puzzle Problem

- We can use the search strategy discussed before, e.g. BFS, A*



8-puzzle Problem

- Let's trace a case with A* algorithm

Given an initial state of a 8-puzzle problem and final state to be reached-

2	8	3
1	6	4
7		5

Initial State

1	2	3
8		4
7	6	5

Final State

2	8	3
1	6	4
7		5

Initial State

1	2	3
8		4
7	6	5

Final State

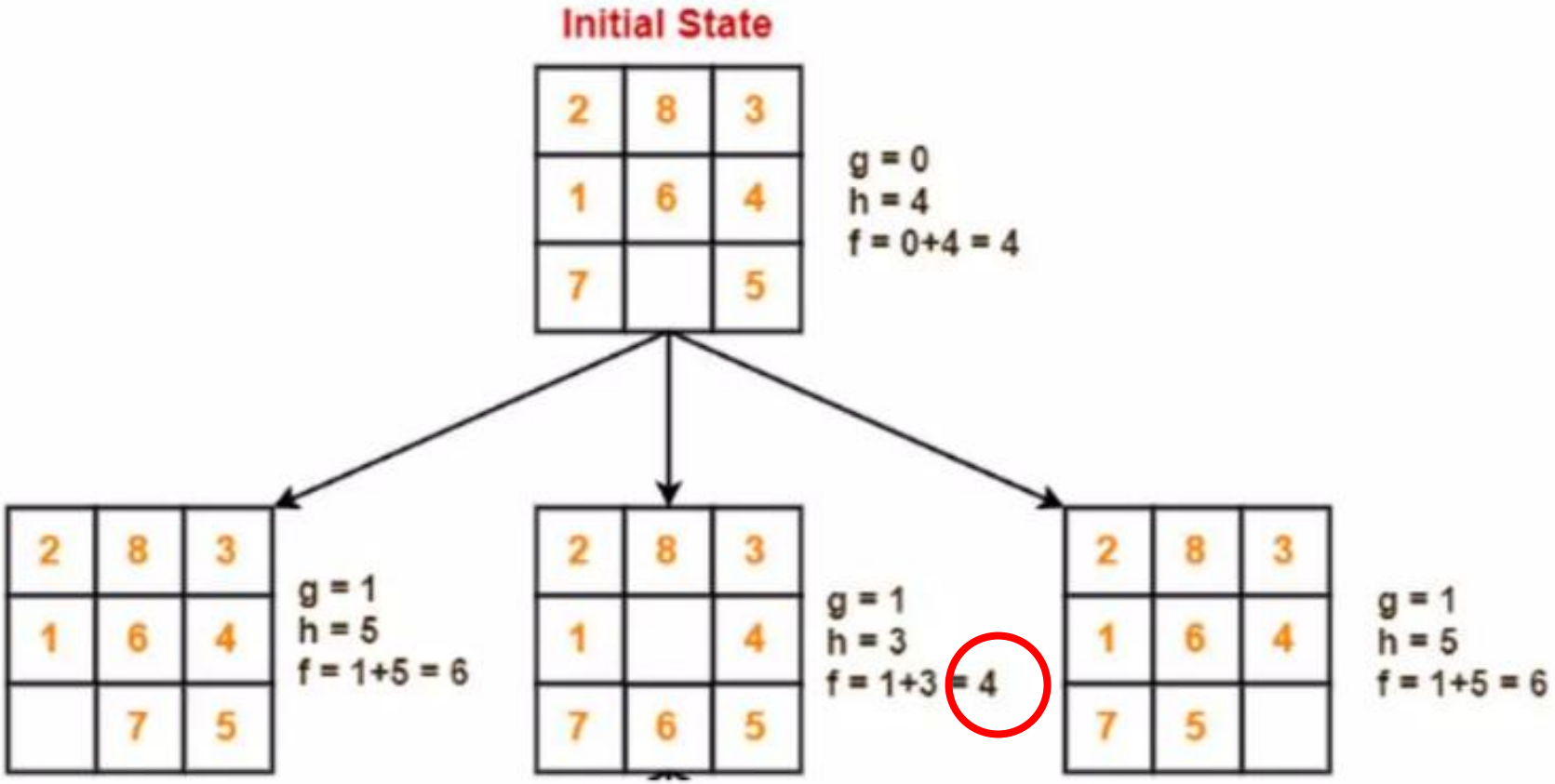
Initial State

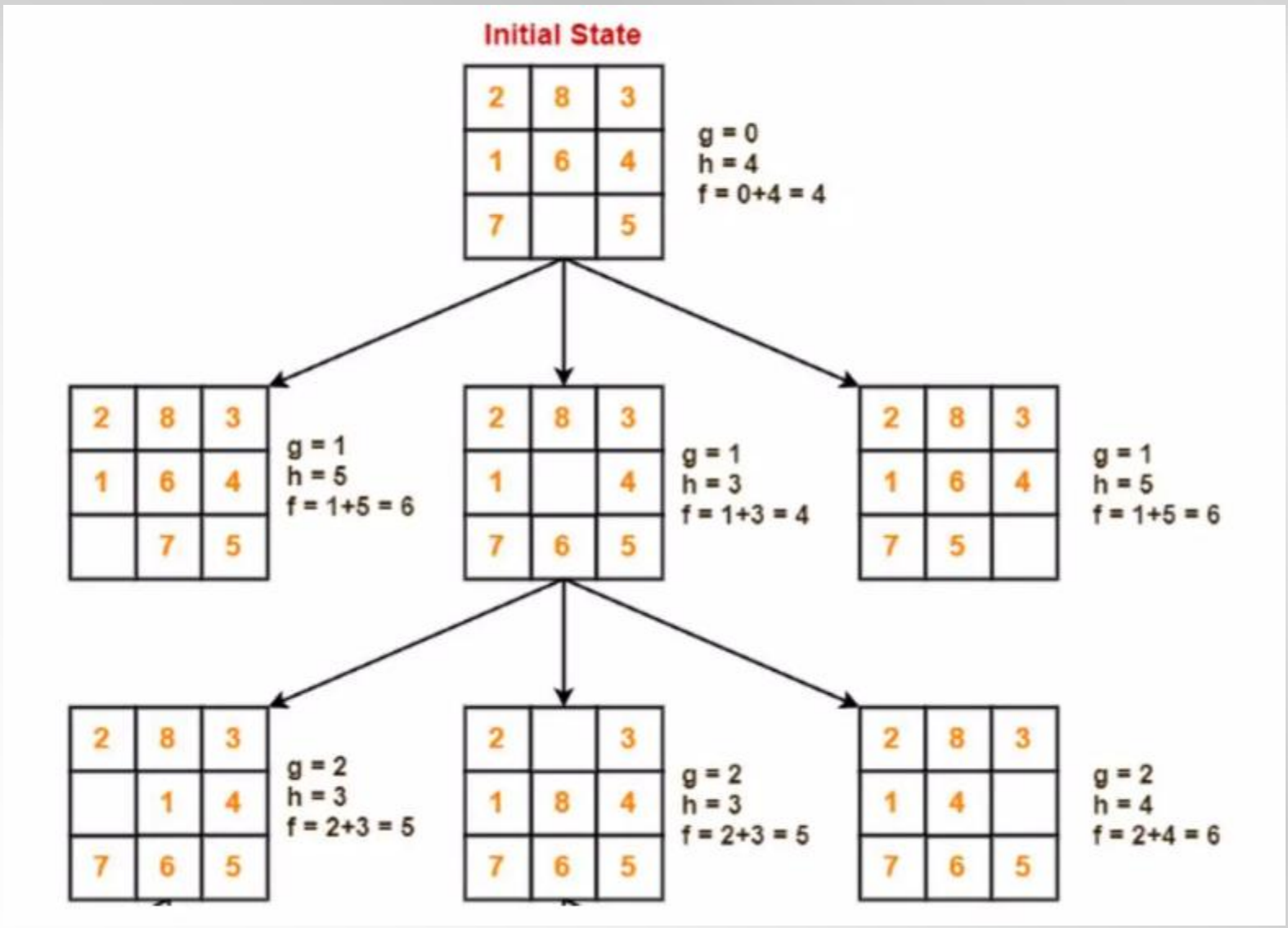
2	8	3
1	6	4
7		5

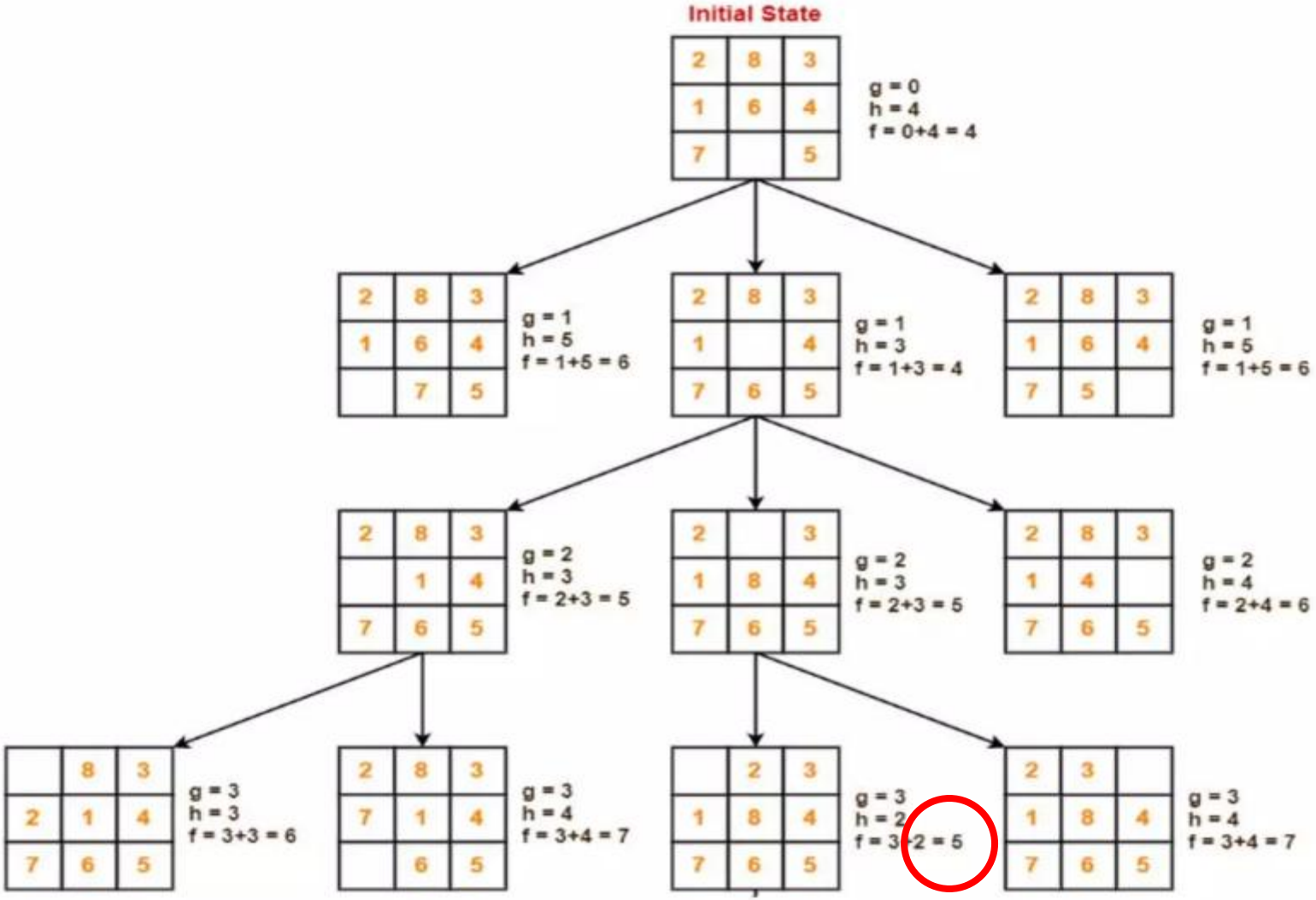
g = 0
h = 4
f = 0+4 = 4

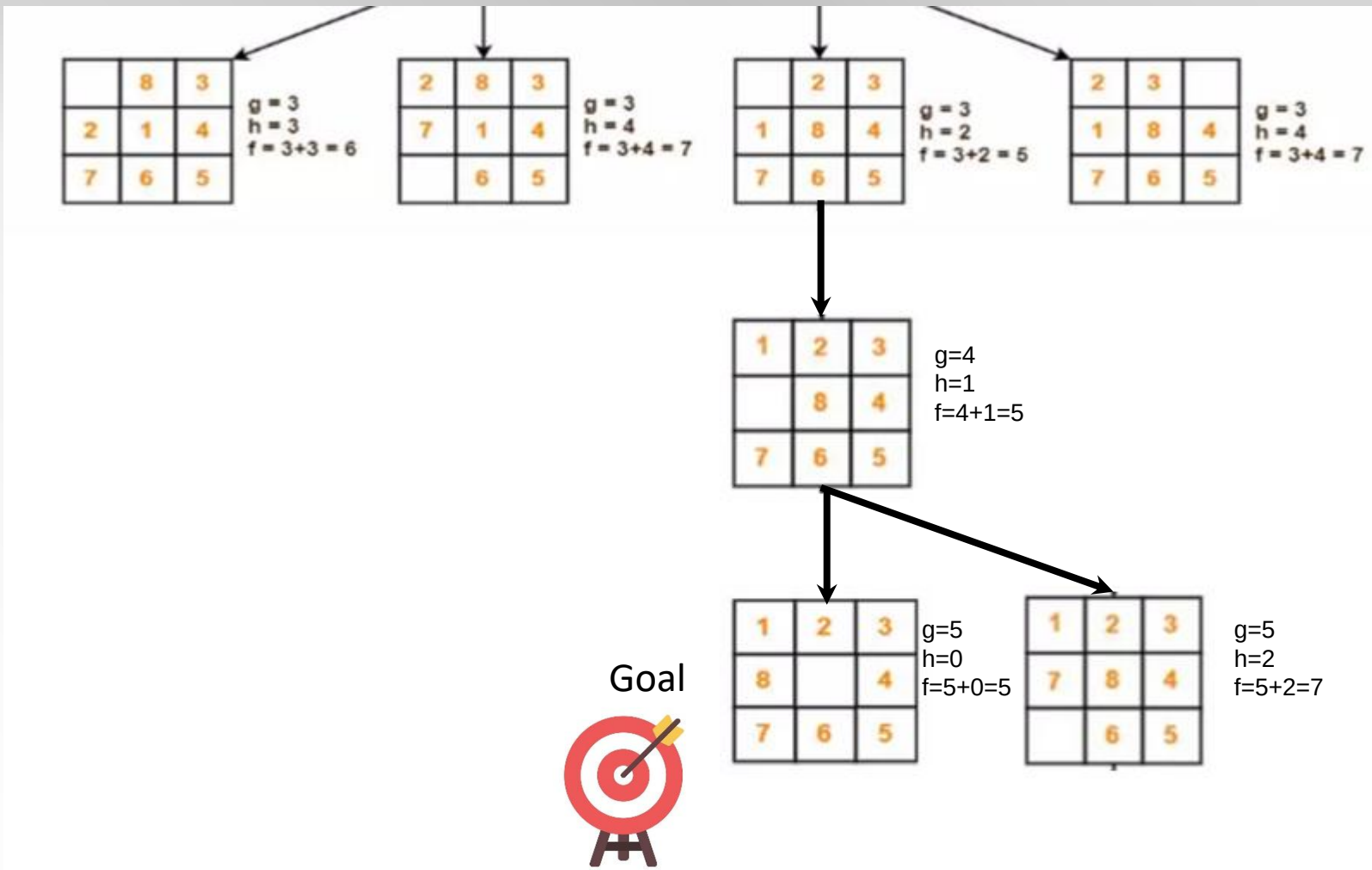
Consider ,
g(n) = Depth of node = no. of moves
and
h(n) = Number of misplaced tiles.

1. Up
2. Down
3. Right or
4. Left


Initial State
2 8 3
1 6 4
7 5
g = 0
h = 4
f = 0+4 = 4
2 8 3
1 6 4
7 5
g = 1
h = 5
f = 1+5 = 6
2 8 3
1 4
7 6 5
g = 1
h = 3
f = 1+3 = 4
2 8 3
1 6 4
7 5
g = 1
h = 5
f = 1+5 = 6

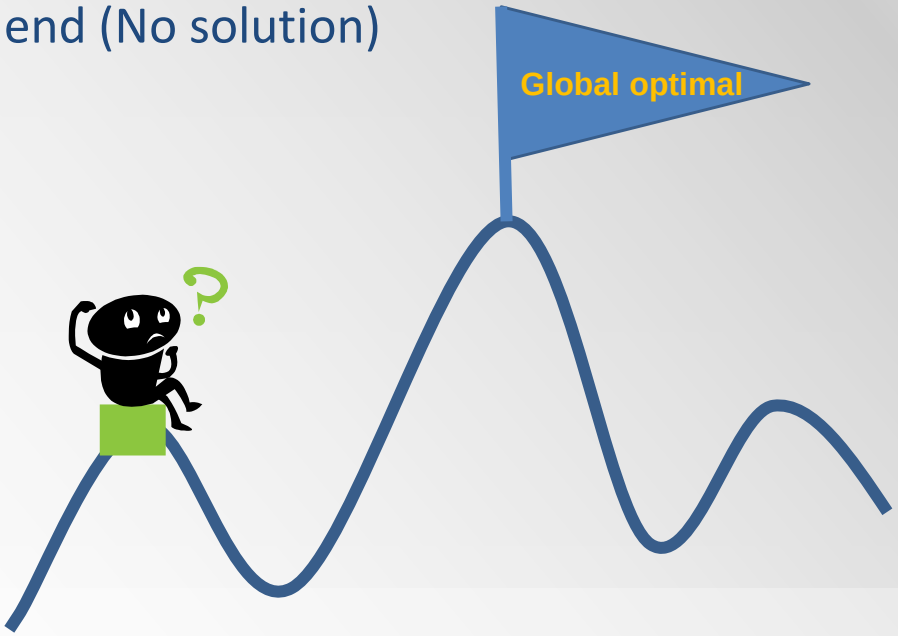

Initial State
2 8 3
1 6 4
7 5
g = 0
h = 4
f = 0+4 = 4
2 8 3
1 6 4
7 5
g = 1
h = 5
f = 1+5 = 6
2 8 3
1 4
7 6 5
g = 1
h = 3
f = 1+3 = 4
2 8 3
1 6 4
7 5
g = 1
h = 5
f = 1+5 = 6
2 8 3
1 4
7 6 5
g = 2
h = 3
f = 2+3 = 5
2 3
1 8 4
7 6 5
g = 2
h = 3
f = 2+3 = 5
2 8 3
1 4
7 6 5
g = 2
h = 4
f = 2+4 = 6


Initial State
g = 0
h = 4
f = 0+4 = 4
g = 1
h = 5
f = 1+5 = 6
g = 1
h = 3
f = 1+3 = 4
g = 1
h = 5
f = 1+5 = 6
g = 2
h = 3
f = 2+3 = 5
g = 2
h = 3
f = 2+3 = 5
g = 2
h = 4
f = 2+4 = 6
g = 3
h = 3
f = 3+3 = 6
g = 3
h = 4
f = 3+4 = 7
g = 3
h = 2
f = 3+2 = 5
g = 3
h = 4
f = 3+4 = 7


g = 3
h = 3
f = 3+3 = 6
g = 3
h = 4
f = 3+4 = 7
g = 3
h = 2
f = 3+2 = 5
g = 3
h = 4
f = 3+4 = 7
g=4
h=1
f=4+1=5
g=5
h=0
f=5+0=5
g=5
h=2
f=5+2=7
Goal

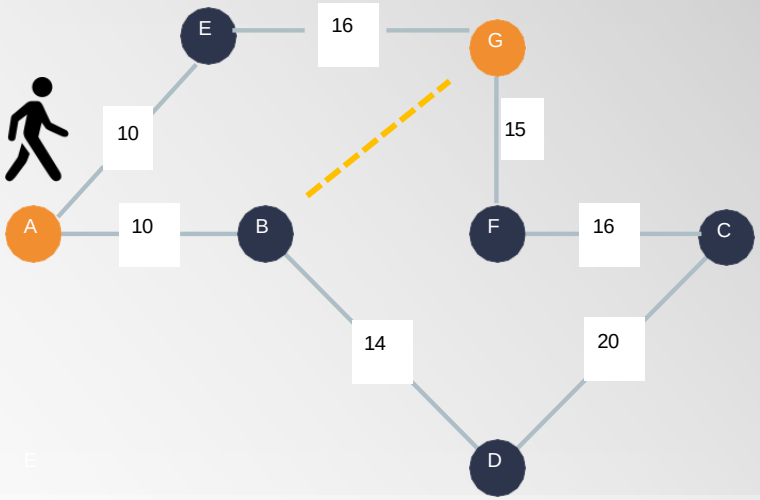
Issues of Heuristic Search

- Heuristics may not always produce the optimum choice
  - Some times it may lead you to local minimum
  - (2nd or 3rd best solution)
  - or even dead end (No solution)



Issues of Heuristic Search

- The definition of a heuristic function h(n) can cause the issue
  - Since it is simply **an estimation**
- If the straight line distance of B->G is 10, it will be attracted to expend node B instead of E

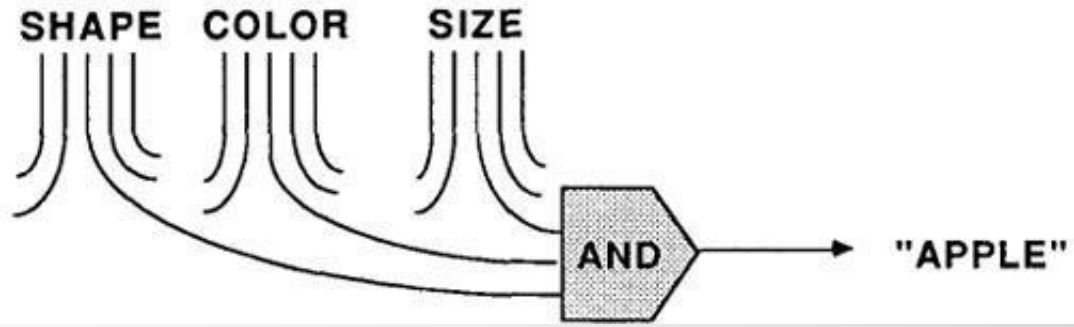


Agenda

- Rational agent
- Search agent
  - Uninformed search and informed search
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  - Heuristics
  - Domination of heuristic functions
- Technological trends of AI agent

Technological Trends of AI Agent

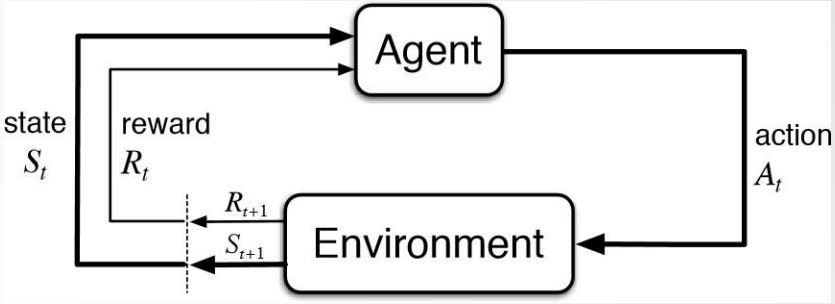
- The evolution of AI agents has undergone several stages
  - Symbolic agent stage: symbolic agent approaches employ logical rules and symbolic representations to encapsulate knowledge and facilitate reasoning processes.
  - A classic example of this approach is knowledge-based expert systems.



Minsky M. Society of mind[M]. Simon and Schuster, 1988.

Technological Trends of AI Agent

- The evolution of AI agents has undergone several stages
  - Learning-based agent stage:
    - Statistical learning from large datasets
    - Utilize ML like neural networks and reinforcement learning methods to train agents to learn through collected data or interactions with their environments, enabling them to achieve maximum cumulative rewards in specific tasks.

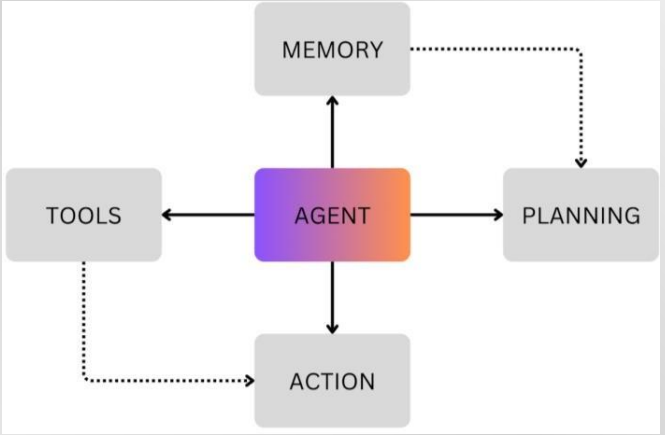


- The agent-environment interaction in reinforcement learning
  - An agent (decision maker) takes the action A_t in the given state s_t and receives the reward R_t.
  - After the reward is received, the agent update their parameters such that the total maximizing discounted expected reward.

Sutton R S, Barto A G. Reinforcement learning: An introduction[M]. MIT press, 2018.

Technological Trends of AI Agent

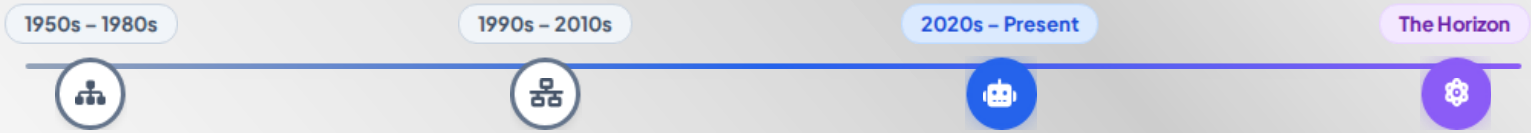
- The evolution of AI agents has undergone several stages
  - Large language model-based agent stage: employ large language model as the primary component of brain or controller of agents.
    - LLMs can exhibit reasoning and planning abilities comparable to symbolic agents
    - LLMs can acquire interactive capabilities with the environment, by learning with feedback
    - LLMs undergo pre-training and demonstrate the capacity for few- and zero-shot generalization
  - Several key components in LLM
    - Memory: The primary function of the memory system is to preserve and regulate knowledge, experiential data, and historical information.
    - Planning: The planning capability defines an LLM-based agent's ability to devise action sequences based on set objectives and existing environment constraints, ensuring goal fulfillment.
    - Tool use: encompassing APIs, calculators, code interpreters, or actions within a physical environment through text-based directives.



https://lilianweng.github.io/posts/2023-06-23-agent/

Evolution to Modern AI Agent Systems

- AI agent architecture has evolved from rigid rule deduction to statistical learning and generative planning, converging toward Neuro-Symbolic AI to deliver both deep perception and auditable logical reasoning.



1. Symbolic Era
CLASSICAL / RULE-BASED

- Hand-crafted IF-THEN rules and logic predicates
- Deterministic deduction engines
- High explainability, brittle execution

2. Connectionist
ML & NEURAL NETS

- Statistical learning from large datasets
- Perception, audio & visual pattern recognition
- Black-box, data-hungry models

3. Generative AI
LLMS & REACT AGENTS

- Foundation models with zero-shot reasoning
- Tool calling & autonomous step planning
- Dynamic working & long-term memory

4. Neuro-Symbolic
HYBRID FRONTIER

- Combines NN perception + Symbolic rule logic
- Verifiable safety & auditability
- Generalizable, trustworthy autonomy

Agentic Workflow



Summary

- Search agent
- Informed Search and Uninformed Search
- Heuristic Search
- A* Search